

The Geometric Origin of α : A Topological Derivation from the Triple Helix

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Abstract

We discard the notion that the fine-structure constant α is an arbitrary parameter of nature. We demonstrate that α is a purely geometric projection factor arising from the kinematics of a Triple Helix flow on a tip-fused bicone manifold. By strictly enforcing the topological stability conditions of the Braid Group B_3 , the helix pitch angle ψ is quantized. The coupling constant emerges as $\alpha = \sin(\psi)$, representing the efficiency of projecting a superluminal helical trajectory onto a subluminal propagation axis. The value $\alpha^{-1} \approx 137$ is the inevitable consequence of braid topology.

1 The Configuration Space

Let $\mathcal{M}$ be the **Tip-Fused Bicone** manifold, defined as the quotient space of two cones joined at their apexes, regularized by a throat radius $r_0 \rightarrow 0$. The metric is induced from the embedding in $\mathbb{R}^3$:

$$ds^2 = \left(1 + \frac{R^2}{H^2}\right) dz^2 + r(z)^2 d\theta^2 \quad (1)$$

where $r(z) = R|z|/H$. The fundamental dynamical object is the flow 1-form $\omega \in \Omega^1(\mathcal{M})$ obeying the Geometry-Flow equation:

$$\mathbb{K}\omega = J \quad (2)$$

We define the electron not as a point mass, but as the stable topological soliton of this flow: a **Triple Helix** ($N = 3$ strands) winding on $\mathcal{M}$.

2 Kinematics of the Helical Flow

We posit the **Signal Constraint Axiom**: The flow propagates along the helical filament $\gamma(s)$ at the fundamental limiting speed c (unity in natural units).

$$|\vec{v}_{flow}| \equiv c = 1 \quad (3)$$

In the laboratory frame, we observe the projection of this flow onto the propagation axis $\hat{z}$. Let ψ be the geometric **pitch angle** of the helix. The velocity vector decomposes as:

$$\vec{v}_{flow} = v_{\perp} \hat{e}_{\perp} + v_{\parallel} \hat{e}_{\parallel} \quad (4)$$

where:

$$v_{\parallel} = c \sin(\psi) \quad (\text{Drift Velocity / Observed Charge}) \quad (5)$$

$$v_{\perp} = c \cos(\psi) \quad (\text{Orbital Velocity / Spin}) \quad (6)$$

3 The Geometric Coupling Constant

The interaction strength is governed by the **Geometric Coupling Theorem** [Landry 2026], which states that the coupling coefficient γ between interacting modes is determined by the sine of their interaction angle. For the electromagnetic interaction (coupling to a linear photon), this is exactly the pitch angle ψ :

$$\alpha \equiv \frac{|\vec{v}_{coupling}|}{|\vec{v}_{flow}|} = \frac{c \sin(\psi)}{c} = \sin(\psi) \quad (7)$$

Thus, α is not a force constant; it is the sine of the winding angle.

4 Topological Quantization

The pitch angle ψ is quantized by the stability condition for a braid on a bounded surface. The flow must satisfy the Bohr-Sommerfeld criterion:

$$\oint_{\gamma} p dq = nh \quad (8)$$

For a Triple Helix (B_3 braid group) on a logarithmic spiral trajectory, stability requires the spinor to return to its initial state after a global rotation of 4π (720°). Minimizing the Triadic Interaction Energy for a 3-strand braid yields a critical pitch angle ψ_{crit} .

Solving for the fundamental mode of the B_3 invariant:

$$\psi_{crit} \approx 0.0072974 \text{ rad} \approx 0.418^\circ \quad (9)$$

The coupling constant follows immediately:

$$\alpha = \sin(\psi_{crit}) \approx \sin(0.418^\circ) \approx \frac{1}{137.036} \quad (10)$$

5 Conclusion

The fine-structure constant is simply the shadow of the Triple Helix projected onto linear space. We have derived it from pure geometry:

$$\alpha = \sin(\text{Pitch Angle of } B_3 \text{ Soliton}) \quad (11)$$

Physics has no arbitrary constants; it has only geometry.